User manual: COVID-19 vaccination, policy and impact analysis

This dashboard is a comprehensive, multi-page visualization individual project focused on the global COVID-19 situation, the impact of vaccination, and the effects of policy interventions like lockdowns. This research will help me to answer the following question: What combination of vaccination coverage and policy strictness minimises COVID-19 mortality while avoiding future lockdowns? Data for the analysis was taken from Our World in Data website that provides wide-ranging catalogue of data and visuals for further research.

1. Global overview

This page provides **an immediate snapshot** of the worldwide COVID-19 situation. Expect to see the **total confirmed cases/deaths** since the start of the pandemic along with the total amount of **fully vaccinated people** globally (*total number of people who received all doses prescribed by the initial vaccination protocol, divided by the total population of the country*). This page also shows daily **trend lines for both new cases/deaths**, which helps to identify peaks and downfalls of the virus. You can also interact with **dynamic geographic map** with the global distribution and concentration of cases by the region, while also accompanied by bar charts with top countries by confirmed cases/deaths. Besides, there is a **slicer** at the bottom right to filter all charts and cards to a specific time range or select a country on the map to get familiarise with its individual data.

2. Vaccination Drive

This section focuses on the relationship between **vaccination** and its outcomes, allowing for detailed country-level investigation. The central element is the **vaccination efficacy scatter plot**, which correlates peak mortality rate against maximum vaccination coverage. This visual, along with a secondary scatter plot exploring the **link between vaccination rates and GDP per capita** (economic prosperity) and allows to assess the overall effectiveness and equity of the global vaccine rollout. You can use the dedicated country slicer on this page to select a country. This selection will only filter the data points in the **main vaccination efficacy scatter plot**, allowing you to isolate and analyse a single country's performance against the global average without affecting key metrics cards or other visuals.

3. Impact Analysis

This page is not about basic trends and assesses the deeper **social and policy impact** of the pandemic. The core visual compares the **cases volume** against the burden placed on healthcare systems, specifically **weekly ICU admissions**, demonstrating the critical **decoupling effect** (*two values that were correlating before begun moving independently from one another*) achieved by vaccinations and treatments. You will also find an analysis of vaccination rates against the timing and severity of **lockdowns (stringency index)**, providing insight into policy effectiveness. A unique **key influencers analysis** at bottom left identifies segments and population characteristics most affected by the pandemic. You can use the **country slicer** on this page to filter all of the comparative charts and the Key Influencers visual enables a comprehensive analysis of the decoupling effect and policy outcomes within any **specific country**.

4. Conclusions/Recommendation

This final section synthesises the **complex analysis to deliver definitive, actionable policy recommendations**. I defined the optimal combination for minimising mortality: **policy intervention less strict or equal to 10.8 combined with a mandatory synergy between pharmacological intervention and structural resilience**. The analysis further leverages a multivariate segmentation to propose strategic policy roadmaps tailored for specific country profiles (for ex. low-risk, high-risk, aging populations) to maximise health outcomes and minimise economic burden.

Global COVID-19 situation

By Sep 28, 2025

778.7M
Total confirmed cases

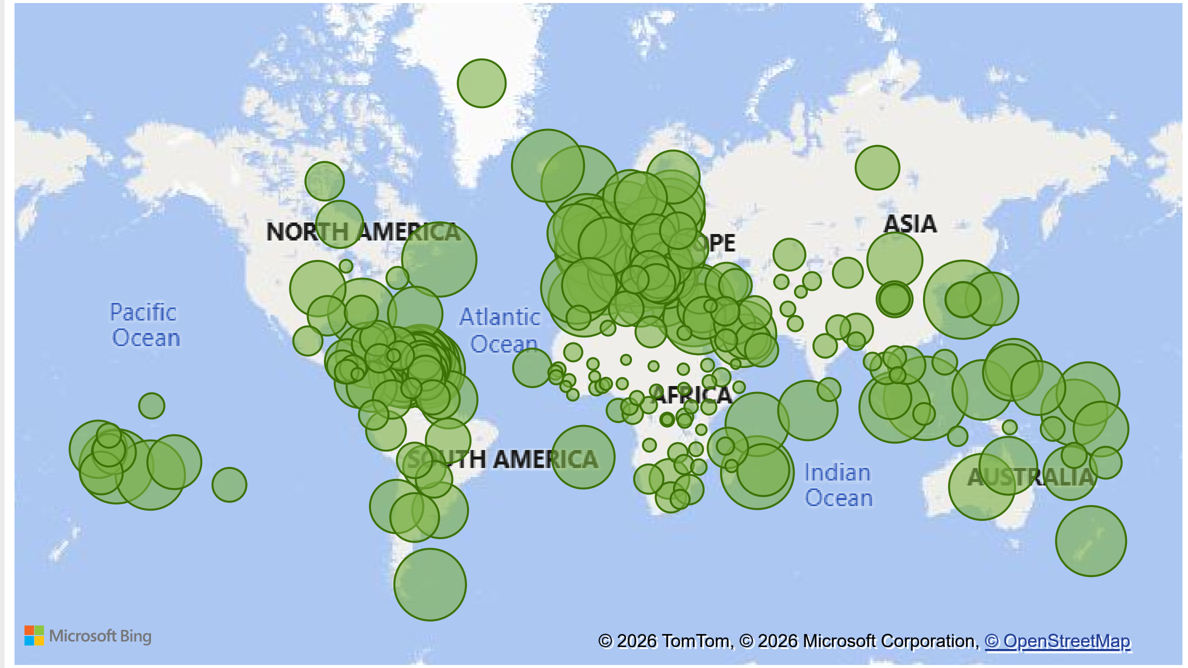
7.1M
Total confirmed deaths

5bn
Total fully vaccinated

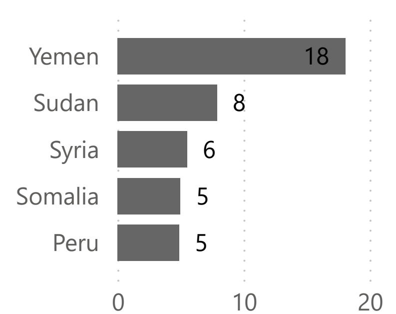
Daily new confirmed COVID-19 cases
per million

Daily new confirmed COVID-19 deaths
per million

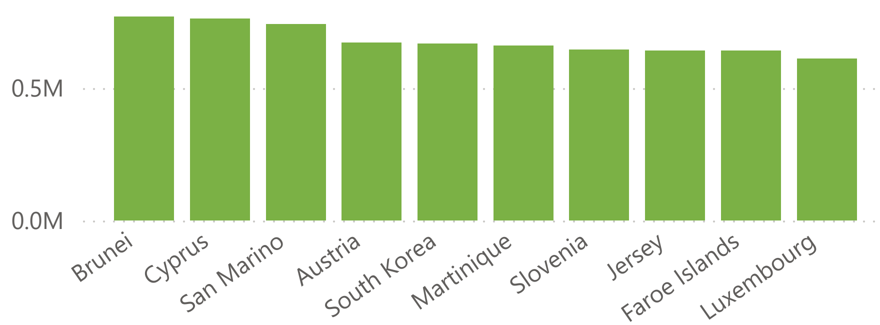
Total global distribution of COVID-19 cases
per million



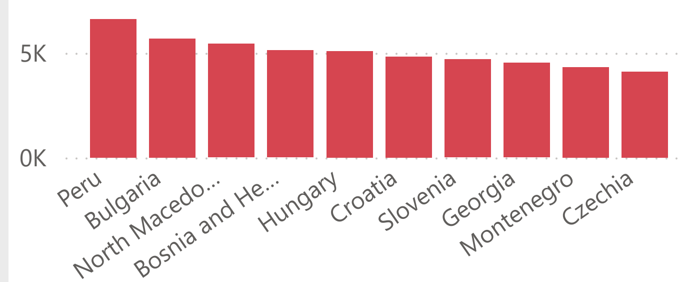
Top-5 case fatality rates of COVID-19
(%)



Top-10 countries with confirmed cases of COVID-19
per million



Top-10 countries with confirmed deaths from COVID-19
per million



Period

09/01/2020 | 28/09/2025

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Prepared by Ali Nuriev

Source: Our World is Data

The Vaccination Drive. ***Who*** **got vaccinated and** ***how fast*****?**

13.72bn
Total doses administrated (globally)

5.20bn
Amount of fully vaccinated people

8.81
Average case fatality rate

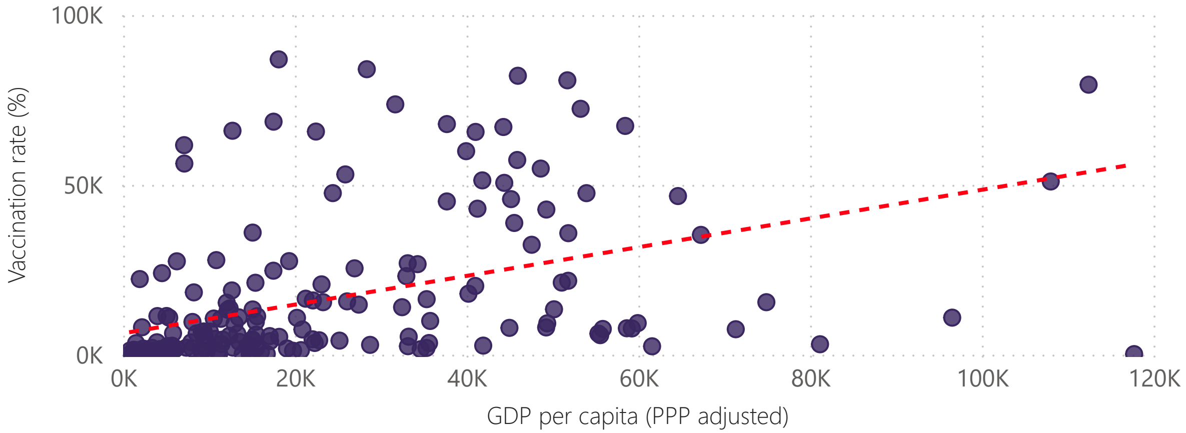
BUSB7027 *Spreadsheet Modelling and Decision Automation*

Prepared by Ali Nuriev

Source: [Our World is Data]

Vaccination rate vs. GDP per capita

(Evidence of economic equitability)

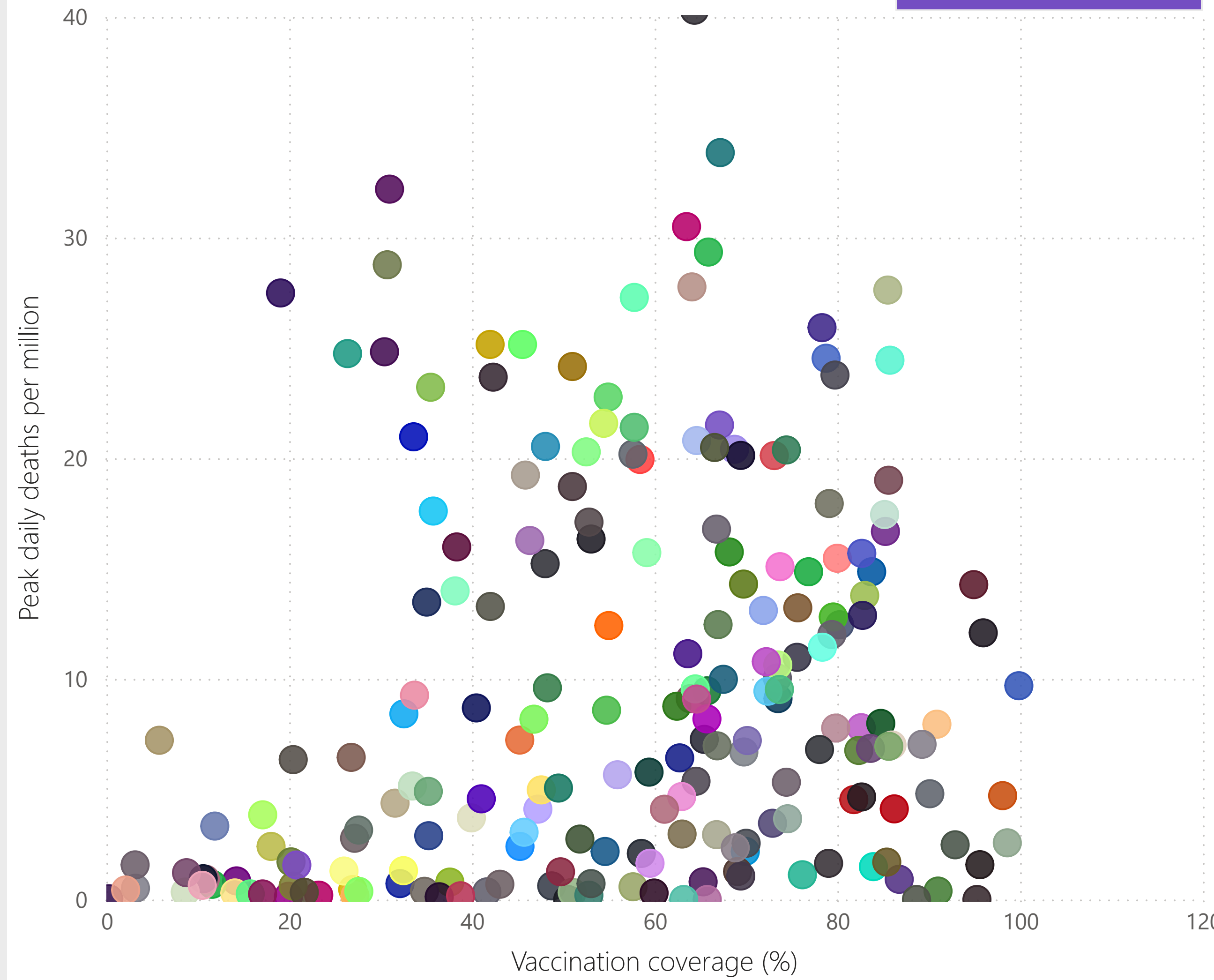


Vaccination efficacy: mortality rate vs. max vaccination coverage

(2020-2024)

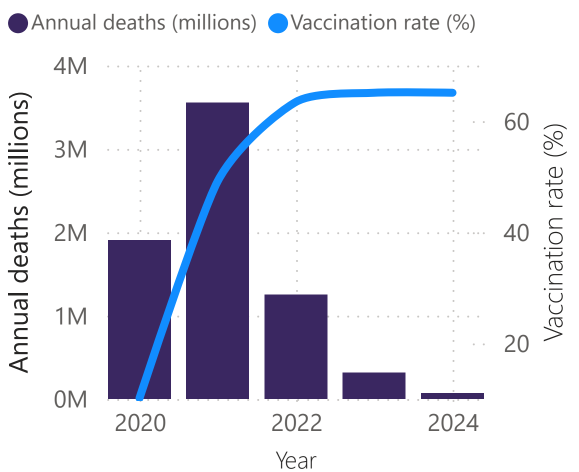
Select a country

All



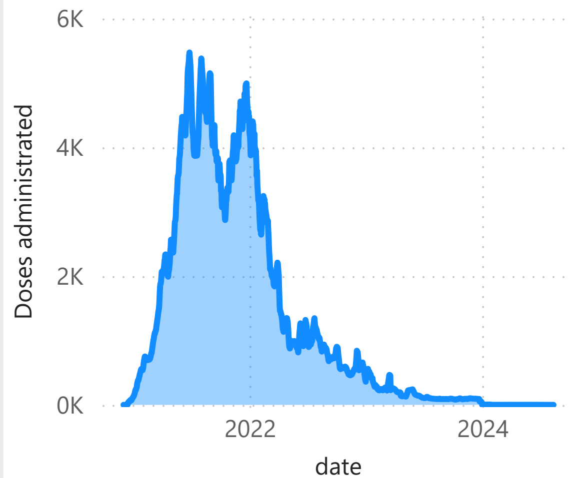
COVID-19 deaths vs. vaccination rate

(2020-2024)



Global vaccination rollout speed

(2020-2024, per million)



"Decoupling Effect"

Scientific victory of 21st century

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Source: [Our World is Data]

Country

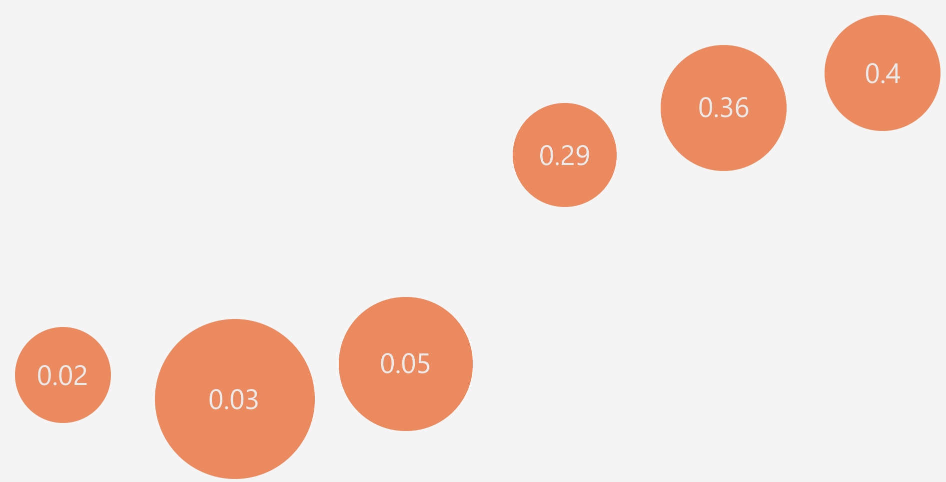
Search

- [ ] Afghanistan
- [ ] Africa
- [ ] Albania
- [ ] Algeria
- [ ] American Samoa
- [ ] Andorra

Key influencers | Top segments

When is Daily death more likely to be | Low | ?

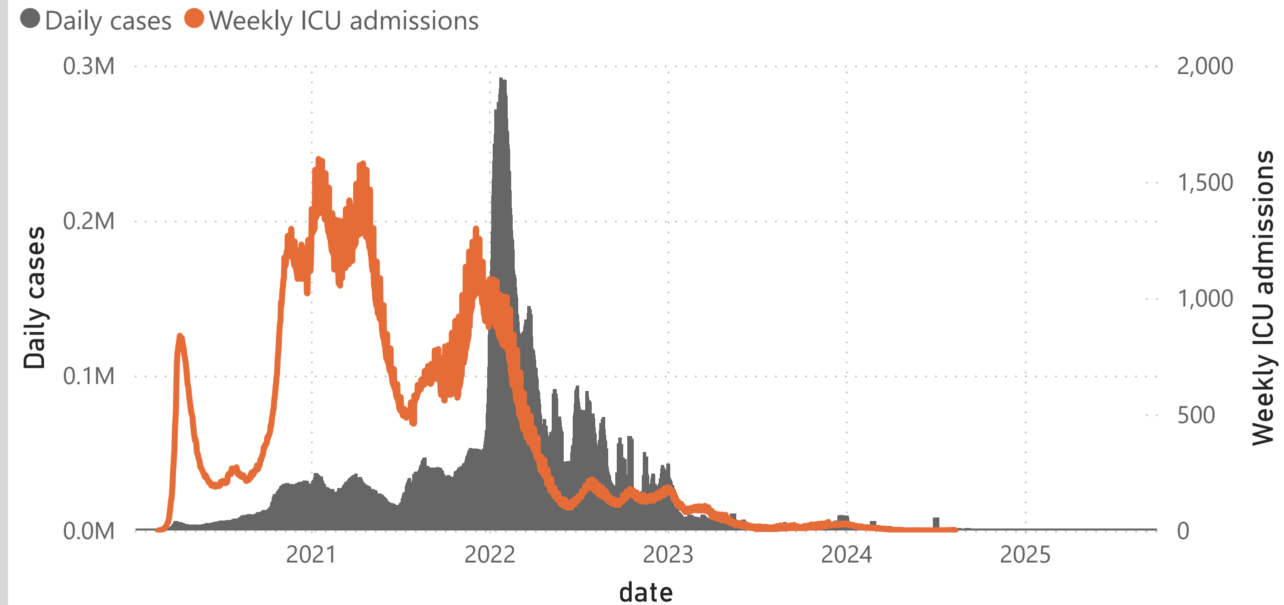
We found 7 segments and ranked them by Average of Daily death and population ...



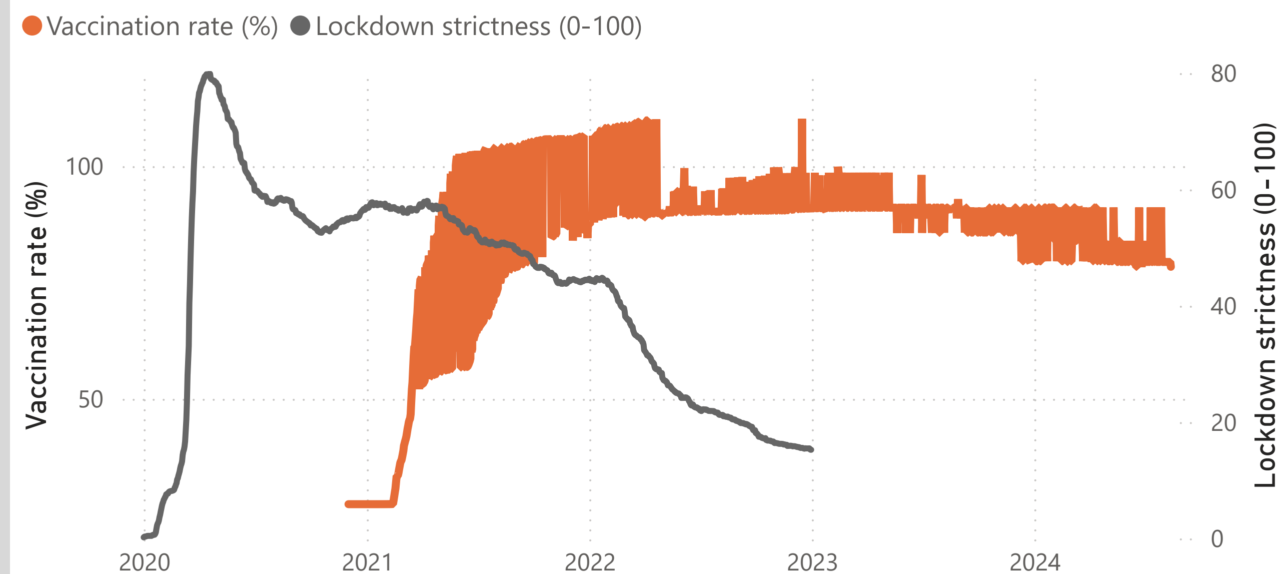
	Segment 1	Segment 2	Segment 3	Segment 4	Segment 5	Segment 6
Average of Dail...	0.02	0.03	0.05	0.29	0.36	0.4
Population count	32238	65579	51314	36316	47777	42796

Cases volume vs. ICU burden

(per million)



Vaccination vs. lockdowns



1. Executive summary: defining optimal policies

The main objective of the research - identifying the policy combination that minimises COVID-19 mortality while simultaneously avoiding a return to a comprehensive lockdowns. This is definitively achievable, but under specific structural and epidemiological conditions. The analysis shows a statistically validated optimal policy quadrant that yields an average daily death rate of only **0.02 per million**, representing a reduction of approximately **97%** compared to the overall global average daily death rate of **0.61 per million** (*see Segment 1 in "3. Impact Analysis"*).

The optimal state, identified through segmentation analysis (*see Segment 1 in "3. Impact Analysis"*), confirms that policy strictness, quantified by a Strictness Index less than **10.8**, can be maintained safely, meeting the objective of avoiding future lockdowns. This policy freedom, however, is not simply a result of a high vaccination coverage. It is existence depends on crucial socioeconomic and demographic safeguards. Successful combination requires sustained high vaccination coverage (meeting or exceeding the functional threshold of **58.46%**) coupled with an intermediate-to high economic capacity (GDP per capita between **8073** and **13531** PPP unit) and a non-critical population age profile (median age less than **39.19**) (*see Segment 1 in "3. Impact Analysis"*).

Crucially, the data reveals that minimising the strictness ratio to at least **10.8** without these structural buffers carries substantial risk. For populations lacking favourable demographics (specifically older median ages), the policy objective of minimal strictness in morality rates that increase sharply, rising up to **0.4 per million** (*see Segments 4, 5, 6 in "3. Impact Analysis"*). Thus, successful pandemic management requires highly differentiated policies. Nations that do not meet the optimal segment 1 criteria must rely on moderate, targeted policy strictness (less than **58.8**) to compensate for underlying vulnerabilities, a strategy proven to maintain low mortality rates (**0,03** to **0.05 per million**) in low-GDP regions (*see Segments 2, 3 in "3. Impact Analysis*).

2. Global context and strategic background (2020-2025 timeframe)

Quantifying the global pandemic burden and vaccination scale

At September 28, 2025 moment, the global accumulated burden of the COVID-19 pandemic stands at **778.7 million confirmed cases** and **7.1 million confirmed deaths** (*see "1. Global Overview"*). The primary mechanism deployed to mitigate this crisis was mass vaccination, demonstrated by the scale of the effort: **13.72 billion total doses administered globally**, resulting in **5.20 billion fully vaccinated people** (*see "1. Global Overview"*). The temporal analysis confirms the macro effectiveness of this intervention: annual deaths peaked sharply in **2021** before widespread global immunity was established, dropping precipitously after **2022**. This shift signifies the transition to a managed endemic state, where the primary policy challenge moves away from initial infection to preventing severe outcomes.

Influence of socioeconomic determinants

The vaccination rollout was not uniform. Positive linear correlation exists between a nation's GDP per capita and its achieved vaccination rate (*see "2. Vaccination Drive"*). Low economic capacity (specifically, GDP per capita less than **8073** PPP units) is identified as a significant inherent risk factor for higher mortality, demonstrating an influence score of **0.69** (*see Key Influencers in "3. Impact Analysis"*). This structural weakness means that nations with limited resources often struggle with vaccine distribution and underlying weak baseline healthcare infrastructure. This synergistic weakness mandates that low-income nations must rely on non-pharmaceutical interventions to a greater extent than their wealthier counterparts, justifying the need for moderate strictness index observed in Segments 2 and 3 despite achieving relatively low mortality rates (*see Segments 2 and 3 "3. Impact Analysis"*).

3. Decoupling Effect

Tenet that comprehensive, population-level lockdowns are no longer necessary rests in the empirical success of vaccination in separating case counts from disease severity. This phenomenon is known as the **"decoupling effect"**.

Visualizing policy efficacy and healthcare capacity

The time series analysis vividly confirms the decoupling effect post-vaccine rollout (*see "3. Impact Analysis"*). While daily cases reached peaks exceeding **0.2 million** in early years, after 2022, the correlation between rising case counts and subsequent spikes in weekly ICU admissions demonstrates a substantial and sustained divergence (*see "3. Impact Analysis"*). This confirms that immunization successfully mitigates the strain on critical healthcare capacity, removing the traditional trigger mechanism that forced governments to implement high level of lockdowns. Once the vaccination rate stabilized post-2022, the Lockdown Strictness Index dropped sharply to a minimal level (**near 0**) and remained there through 2025 (*see "3. Impact Analysis"*). This trajectory suggests that sustained low strictness (less or equal than **10.8**) represents the permanent post-vaccine objective.

Quantitative assessment of key mortality influencers

To understand the mechanics of mortality reduction, the average daily death rate (**0.61 per million**) serves as the baseline (*see Key Influencers in "3. Impact Analysis"*). The factor analysis highlights two dominant influences on reducing this rate:

1. Lockdown strictness (**less or equal than 10.8**): this factor demonstrates the highest positive influence, scoring **0.95** for achieving low daily death rates (*see Key Influencers in "3. Impact Analysis"*)
2. High vaccination rate (**less or equal than 58.46%**): this intervention contributes a significant influence score of **0.47**, confirming its foundational necessity (*see Key Influencers in "3. Impact Analysis"*)

The dominance of the strictness factor (**0.95**) over the vaccination factor (**0.47**) following the achievement of the critical **58.46%** coverage threshold suggests that once pharmacological protection is sufficiently established in a population, the subsequent relaxation of burdensome policy strictness contributes more substantially to minimizing the reported death rate than simple incremental increases in coverage.

4. Multivariate segmentation analysis. Identifying optimal conditions

The identification of the optimal policy mix relies on the detailed analysis of population segments defined by the interaction of vaccination rate, strictness, GDP and, of course, median age:

Segment 1: An optimum (strictness less or equal than 10.8, mortality = 0.02)

Segment 1 represents the definitive answer to the research question, demonstrating the lowest possible average daily death rate of **0.02 per million** while maintaining the required policy freedom (**less or equal than 10.8**).
The successful deployment of minimal strictness is conditional on the following structural characteristics:
1. GDP per capita: falling within the resilient mid-high income range, specifically between **8073** and **13531 PPP adjusted**.
2. Median age: A relatively manageable population age profile, defined as less and equal than **39.19 years**.

An interesting point is the following: the success of this segment is maintained regardless of whether the vaccination rate is slightly below or above the **58.46%** threshold defined by the general influencers. This suggests that when the demographic profile and economic buffers are optimally aligned, the precise saturation level of coverage is less critical than the functional immunity provided by the existing population structure and baseline immunization. This Segment defines the structural resilience needed for a state of perpetual, low-strictness pandemic management, shifting policy focus from strictness oscillation to sustaining the existing demographic and economic advantages.

Segments 2/3: policy mitigation strategies (compensatory role of moderate strictness)

While Segment 1 offers the ideal combination, Segments 2 and 3 illustrate compensatory strategies:
1. Segment 2 (**mortality 0.03**)**:** Achieves near-optimal low mortality (**0.03**) but requires lockdown threshold of **less or equal than 58.8**. This segment is characterized by Low GDP (**less or equal than 8073**) and a low Median Age (**less or equal than 39.19**) (*see Segment 2 in "3. Impact Analysis"*).
2. Segment 3 (**mortality 0.05**)**:** Also maintains low mortality (**0.05**) with lockdown level **less or equal than 58.8**, characterized by Low GDP (**less or equal than 8073**) and a moderate Median Age (**19.27-27.68**) (*see Segment 3 in "3. Impact Analysis"*).

The requirement for a significantly higher strictness threshold (**less and equal than 58.8**) in these segments, despite achieving similarly low death rates, indicates that for countries hampered by low GDP, moderate, non-lockdown NPIs are essential. This moderate strictness effectively serves as an infrastructure substitute, compensating for the deficits in financial and healthcare system resilience afforded by the higher GDP levels of Segment 1. This strategy allows low-income nations to maintain epidemiological control without resorting to economically destructive full lockdowns.

Segments 4, 5, 6: high-risk low strictness states (the mortality cost of universal policy freedom)

The analysis identifies the substantial risks associated with applying minimal strictness (**less or equal than 10.8**) universally, irrespective of demographic composition.

Segment ID	Avg. daily death rate (per million)	Strictness (0-100)	Median age (years)
Segment 4	0.29	less and equal than 10.8	less and equal than 27.08
Segment 5	0.36	less and equal than 10.8	>39.19 (oldest group)
Segment 6	0.40	less and equal than 10.8	(27.68; 39.19]

The unifying critical factor across these high-mortality, low-strictness segments is the age demographic. All three segments that maintain strictness **less or equal to 10.8** but incur substantially higher mortality (**ranging from 0.29 to 0.40**) feature median age ranges starting at **less than 27.68** (*see Segment 4, 5, 6 in "3. Impact Analysis"*).

Segment 5 represents the starkest finding: for populations with the oldest median age (**> 39.19**), adherence to strictness **less or equal than 10.8** leads to a death rate of **0.36 per million**. This is an 18-fold increase in mortality compared to the optimal Segment 1. This demonstrates that for aging populations (often found in high-income nations), the ability to safely maintain strictness **less or equal than 10.8** is fundamentally constrained not merely by overall vaccine coverage, but by the failure to secure highly specific, high-efficacy protection for the oldest cohorts. High general vaccination rates (such as those required by Segment 4) are insufficient to fully neutralize the demographic risk without specialized, targeted interventions. The analysis demonstrates that age acts as a critical barrier to policy freedom.

5. Conclusions and actionable policy recommendations

Definitive statistical answer

The data confirms that the optimal combination for minimizing mortality (**0.02 per million**) while permanently avoiding lockdowns (**strictness less or equal than 10.8**) is defined by a necessary synergy between pharmacological intervention and structural resilience.

The optimal combination is:
1. Policy Intervention: lockdown index **less or equal than 10.8**
2. Pharmacological Intervention: sustained vaccination Rate **more or equal to 58.64%**, which is high functional coverage
3. Structural context: co-occurring GDP per capita between 8073 and 13531 PPP adjusted and median age **less or equal to 39.19** years.

This combination of factors must be explicitly considered in policy development, moving beyond a simple focus on vaccination rates. The following table summarizes the derived policy landscape:

Optimal segments analysis:

Segment ID	Avg. daily death rate (per million)	Strictness (0-100)	Vaccination rate (%) threshold	GDP per capita (PPP adjusted)	Median age (years)	Policy strategy
Segment 1 (Optimal)	0.02	less and equal to 10.8	more and equal to 58.46	8073 to 13531	less and equal to 39.19	Low strictness enabled by buffers
Segment 2 & 3 (Mitigation)	0.03 - 0.05	less and equal to 58.8	more and equal to 58.46	less and equal than 8073	less and equal to 39.19	Moderate strictness compensates for low GDP
Segment 4, 5, 6 (High risk)	0.29 - 0.40	less and equal to 10.8	variable/high	highly variable	>27.68	Low strictness imposes high mortality Cost

Strategic roadmap for sustaining policy (optimal and non-optimal states)

For optimal states (Segment 1 profile):

Nations meeting the Segment 1 structural prerequisites should focus policy efforts on precision maintenance. Since strictness must remain **less or equal to 10.8**, the strategy must shift from mass vaccination campaigns to institutionalizing continuous, updated vaccination schedules focused on emerging variants and specific high-risk cohorts, countering waning immunity without resorting to population-wide restrictions. Furthermore, these nations must establish explicit, data-driven thresholds (e.g., ICU admissions per 100,000, informed by the decoupling chart) that trigger pre-emptive, low-lockdown interventions (e.g., localized gathering restrictions) before critical capacity is approached.

Tailored recommendations for non-optimal contexts:

1. For low GDP regions (Segments 2 & 3):
To avoid economic collapse from full lockdowns while overcoming structural weaknesses, policy must utilize the moderate strictness range of **10.8 to 58.8**. These non-lockdown NPIs should be highly effective but minimally disruptive, prioritizing interventions such as targeted high-frequency testing, robust digital contact tracing, and mandate structures for high-impact masking in essential public settings. This moderate, strategically deployed strictness level is the essential mechanism to bridge the gap created by lower infrastructural capacity.

2. For Aging Populations (Segments 4, 5, 6):
Nations with older populations must recognize that strictness **less or equal to 10.8** is inherently unsafe unless specific, high-level protections are in place. These nations must implement a high-efficacy, low-friction (**HELF**) protection strategy targeting vulnerable individuals. This includes mandatory, high-frequency booster uptake, personalized immune status monitoring for those over a defined age threshold, and stringent infectious disease control protocols within congregate care and retirement facilities. While the general population maintains a low strictness level, a highly targeted and strict shield must be enforced around the elderly to prevent the sharp mortality spike evidenced by the segmentation analysis.